

Submission 1 — random forest 3 test, Version 2 Local pAUC@0.01 score 0.918449697286374, Leaderboard pAUC@0.01 score 0.9499202 Description: Baseline Random Forest model with 250 trees, max depth 12, min samples leaf 3, and 5-fold CV.

Submission 2 — random forest 3 test, Version 3 Local pAUC@0.01 score 0.918115845609035, Leaderboard pAUC@0.01 score 0.9503978 Description: Increased number of trees from 250 to 400. Local score dropped slightly but leaderboard improved.

Submission 3 — random forest 3 test, Version 4 Local pAUC@0.01 score 0.9168511369019388, Leaderboard pAUC@0.01 score 0.9509108 Description: Increased max depth from 12 to 14. Leaderboard continues to improve.

Submission 4 — random forest 3 test, Version 5 Local pAUC@0.01 score 0.9167999692228029, Leaderboard pAUC@0.01 score 0.9515123 Description: Reduced min samples leaf from 3 to 2. Another leaderboard gain.

Submission 5 — random forest 3 test, Version 6 Local pAUC@0.01 score 0.9167999692228029, Leaderboard pAUC@0.01 score 0.9515123 Description: Explicitly set max_features to sqrt. No change in scores (was already the default).

Submission 6 — random forest 3 test, Version 7 Local pAUC@0.01 score 0.9171054533267353, Leaderboard pAUC@0.01 score 0.9516008 Description: Added min_samples_split=6. Small improvement on both local and leaderboard.

Submission 7 — random forest 3 test, Version 8 Local pAUC@0.01 score 0.920432844983937, Leaderboard pAUC@0.01 score 0.9502563 Description: Increased trees to 600 and max depth to 16. Big local CV jump but leaderboard dropped, suggesting overfitting.

Submission 8 — random forest 3 test, Version 9 Local pAUC@0.01 score 0.9203335633865966, Leaderboard pAUC@0.01 score 0.9497964 Description: Increased trees to 1000, kept depth 16. Similar local score but leaderboard dropped further.

Submission 9 — random forest 3 test, Version 16 Local pAUC@0.01 score 0.9205319105596038, Leaderboard pAUC@0.01 score 0.9496979 Description: Increased trees to 2000 and reduced min samples leaf to 1. Highest local CV so far but leaderboard still declining.

Submission 10 — random forest 3 test, Version 17 Local pAUC@0.01 score 0.9199850909696234, Leaderboard pAUC@0.01 score 0.9497245 Description: Kept 2000 trees but increased min samples leaf back to 3 to regularize. Slight leaderboard recovery.

Submission 11 — random forest 3 test, Version 18 Local pAUC@0.01 score 0.9183491097398251, Leaderboard pAUC@0.01 score 0.9490269 Description: Tried 1000 trees, depth 18, min samples leaf 5. Worse on both metrics, too much regularization on leaf size.

Submission 12 — random forest 3 test, Version 19 Local pAUC@0.01 score 0.9164249850549824, Leaderboard pAUC@0.01 score 0.9510966 Description: Switched to 600 trees, depth 15, and changed max_features from sqrt to log2. Lower local but decent leaderboard.

Submission 13 — random forest 3 test, Version 20 Local pAUC@0.01 score 0.9176844601460781, Leaderboard pAUC@0.01 score 0.9491418 Description: First ensemble attempt. 50/50 blend of Random Forest and ExtraTrees, both with same hyperparameters (600 trees, depth 15, log2 features). Leaderboard worse than single RF.

Submission 14 — random forest 3 test, Version 21 Local pAUC@0.01 score 0.917519714162478, Leaderboard pAUC@0.01 score 0.9496283 Description: Changed RF to previous best config (400 trees, depth 14, sqrt features) while keeping ET at 600 trees with log2 features. Still 50/50 blend. Slight leaderboard improvement over sub 13.

Submission 15 — random forest 3 test, Version 22 Local pAUC@0.01 score 0.9169640573222151, Leaderboard pAUC@0.01 score 0.9510258 Description: Replaced ExtraTrees with Logistic Regression (with StandardScaler) in an 80/20 blend with RF. Leaderboard improved over ET blends.

Submission 16 — random forest 3 test, Version 23 Local pAUC@0.01 score 0.9172979089995543, Leaderboard pAUC@0.01 score 0.9513619 Description: RF + ET blend (80/20), both with identical params (400 trees, depth 14, sqrt features). Better than 50/50 blend.

Submission 17 — random forest 3 test, Version 27 Local pAUC@0.01 score 0.9172340844141806, Leaderboard pAUC@0.01 score 0.9514946 Description: Same models as sub 16 but shifted blend to 90/10 favoring RF. Small leaderboard gain.

Submission 18 — random forest 3 test, Version 28 Local pAUC@0.01 score 0.9171682959954108, Leaderboard pAUC@0.01 score 0.9515742 Description: Same models, pushed blend to 95/5 RF/ET. Another tiny leaderboard gain, approaching single-RF territory.

Submission 19 — random forest 3 test, Version 29 Local pAUC@0.01 score 0.9186469545318465, Leaderboard pAUC@0.01 score 0.9518838 Description: Kept RF at depth 14, but made ExtraTrees fully unconstrained (depth=None, leaf=1, split=2). 90/10 blend. Notable improvement on both metrics.

Submission 20 — random forest 3 test, Version 30 Local pAUC@0.01 score 0.9177310422742337, Leaderboard pAUC@0.01 score 0.9522818 Description: Reduced RF trees from 400 to 300, kept unconstrained ET. 90/10 blend. Leaderboard jumped to 0.9523.

Submission 21 — random forest 3 test, Version 31 Local pAUC@0.01 score 0.9177741385981113, Leaderboard pAUC@0.01 score 0.9521491 Description: Also reduced ET trees from 400 to 300. 90/10 blend. Slightly worse than sub 20, so 400 ET trees is better.

Submission 22 — random forest 3 test, Version 32 Local pAUC@0.01 score 0.9183328491993055, Leaderboard pAUC@0.01 score 0.9526710 Description: Reduced RF trees further to 200, restored ET to 400 trees. 90/10 blend. New best leaderboard score (0.9527).

Submission 23 — random forest 3 test, Version 33 Local pAUC@0.01 score 0.9184182759520363, Leaderboard pAUC@0.01 score 0.9526179 Description: Same as sub 22 but reduced ET trees to 200. Slightly worse leaderboard, so 400 ET trees confirmed better.

Submission 24 — random forest 3 test, Version 34 Local pAUC@0.01 score 0.9180115464173675, Leaderboard pAUC@0.01 score 0.9508224 Description: Changed RF max depth from 14 to 15, kept 200 RF + 400 ET with 90/10 blend. Leaderboard dropped significantly.

Submission 25 — random forest 3 test, Version 35 Local pAUC@0.01 score 0.920793974307148, Leaderboard pAUC@0.01 score 0.9511585 Description: Changed RF max depth to 16. High local CV but leaderboard still below depth-14 runs.

Submission 26 — random forest 3 test, Version 36 Local pAUC@0.01 score 0.9178232344330142, Leaderboard pAUC@0.01 score 0.9525826 Description: Returned RF to depth 14. Shifted blend to 95/5 RF/ET. Close to sub 22 but slightly worse.

Submission 27 — random forest 3 test, Version 37 Local pAUC@0.01 score 0.9168059687338281, Leaderboard pAUC@0.01 score 0.9496891 Description: Added row-level feature engineering (mean, std, min, max, range, sum, median) to same 90/10 RF+ET setup. Hurt both scores significantly.

Submission 28 — random forest 3 test, Version 38 Local pAUC@0.01 score 0.9188727953723994, Leaderboard pAUC@0.01 score 0.9527860 Description: Three-model ensemble: RF1 (depth 14, sqrt features) + RF2 (unconstrained, log2 features) + ET (unconstrained, sqrt features). Weighted 75/15/10. New best leaderboard (0.9528).

Submission 29 — random forest 3 test, Version 39 Local pAUC@0.01 score 0.9202382094560482, Leaderboard pAUC@0.01 score 0.9502817 Description: Same three models and weights as sub 28 but switched to 10-fold CV. Higher local CV but much worse leaderboard.

Submission 30 — random forest 3 test, Version 40 Local pAUC@0.01 score 0.9185723288627943, Leaderboard pAUC@0.01 score 0.9525737 Description: Three-model ensemble with RF2 changed to match RF1's params (depth 14, sqrt). Weights 80/10/10. Slightly below sub 28.

Submission 31 — random forest 3 test, Version 41 Local pAUC@0.01 score 0.919829182236306, Leaderboard pAUC@0.01 score 0.9527595 Description: Same three models as sub 28 but reweighted to 75/5/20, giving more weight to ExtraTrees. Very close to best.

Submission 32 — random forest 3 test, Version 42 Local pAUC@0.01 score 0.9183083012818538, Leaderboard pAUC@0.01 score 0.9527860 Description: Same three models as sub 28 but reweighted to 75/20/5, giving more weight to RF2 over ET. Tied best leaderboard (0.9528).

Submission 33 — random forest 3 test, Version 43 Local pAUC@0.01 score 0.9192743993019044, Leaderboard pAUC@0.01 score 0.9527418 Description: Same three models, reweighted to 80/5/15. Close to best but slightly below.

Submission 34 — random forest 3 test, Version 44 Local pAUC@0.01 score 0.9196415281361405, Leaderboard pAUC@0.01 score 0.9526268 Description: Same as sub 33 but reduced ET trees from 400 to 200. Slightly worse leaderboard.

Submission 35 — random forest 3 test, Version 45 Local pAUC@0.01 score 0.9188727953723994, Leaderboard pAUC@0.01 score 0.9527860 Description: Repeat of sub 28 config (75/15/10 blend, same models). Confirms reproducibility, tied best leaderboard.

Submission 36 — random forest 3 test, Version 47 Local pAUC@0.01 score 0.9284618207172699, Leaderboard pAUC@0.01 score 0.9491130 Description: Four-model rank-blended ensemble (RF1 + RF2 + RF3-shallow + ET) with balanced_subsample class weights, max_samples=0.8, and rank blending (40/20/20/20). Highest local CV by far but poor leaderboard, rank blending didn't generalize.

Submission 37 — random forest 3 test, Version 48 Local pAUC@0.01 score 0.9227167832236154, Leaderboard pAUC@0.01 score 0.9436633 Description: Same three-model setup as sub 28 but added class_weight=balanced_subsample to RF1 only. Worst leaderboard score of all submissions, balanced class weights hurt badly.

Submission 38 — random forest 3 test, Version 49 Local pAUC@0.01 score 0.919299821125217, Leaderboard pAUC@0.01 score 0.9522641 Description: Same three-model 75/15/10 blend but increased RF1 max depth from 14 to 18. Slightly worse than depth 14 on leaderboard.

Submission 39 — random forest 3 test, Version 50 Local pAUC@0.01 score 0.9174143250432758, Leaderboard pAUC@0.01 score 0.9522907 Description: Same three-model 75/20/5 blend but reduced RF1 max depth to 10. Lower local CV but leaderboard comparable to depth-18 run.

Submission 40 — random forest 3 test, Version 51 Local pAUC@0.01 score 0.9167137765750477, Leaderboard pAUC@0.01 score 0.9493795 Description: Switched from 5-fold to 3-fold CV with three-model 75/15/10 blend. Also changed RF2 to depth 16 (from None). Worse on both metrics, 5-fold is better.

Submission 41 — random forest 3 test, Version 52 Local pAUC@0.01 score 0.9162129990590391, Leaderboard pAUC@0.01 score 0.9493353 Description: Same 3-fold

setup as sub 40 but capped ET at depth 18 (from None) and restored RF2 to unconstrained. Worst local CV, confirms 3-fold is inferior.

Submission 42 — random forest 3 test, Version 52 (resubmit) Local pAUC@0.01 score 0.9186705205325998, Leaderboard pAUC@0.01 score 0.9493353 Description: Two-model RF+ET blend (88/12) instead of three-model ensemble. RF(200, depth=14) + ET(400, depth=None). Simpler than sub 28 but worse leaderboard.

Submission 43 — random forest 3 test, Version 53 Local pAUC@0.01 score 0.9186705205325998, Leaderboard pAUC@0.01 score 0.9527152 Description: Same code as sub 42 (88/12 RF+ET blend) but different submission. Leaderboard much better than sub 42 despite identical local score — highlights local/leaderboard instability.

Submission 44 — random forest 3 test, Version 54 Local pAUC@0.01 score 0.917493, Leaderboard pAUC@0.01 score 0.9535997 Description: First LightGBM submission. Three-model blend: RF(200, depth=14) + ET(400, depth=None) + LGB(300 trees, lr=0.05, depth=4, leaves=15). Weights 75/10/15 (sub 28 style with LGB replacing RF2). New best leaderboard by a large margin.

Submission 45 — random forest 3 test, Version 55 Local pAUC@0.01 score 0.919395, Leaderboard pAUC@0.01 score 0.9525383 Description: Same three-model blend as sub 44 but reduced LGB learning rate from 0.05 to 0.01. Leaderboard dropped significantly, slower learning hurt here.

Submission 46 — random forest 3 test, Version 56 Local pAUC@0.01 score 0.916141, Leaderboard pAUC@0.01 score 0.9458880 Description: Same blend as sub 44 but reduced LGB num_leaves from 15 to 7. Worst leaderboard of the LGB runs, too few leaves.

Submission 47 — random forest 3 test, Version 57 Local pAUC@0.01 score 0.917967, Leaderboard pAUC@0.01 score 0.9509728 Description: Same blend as sub 44 but increased LGB num_leaves from 15 to 31 (LGB default). Worse than 15 leaves, more complexity didn't help.

Submission 48 — random forest 3 test, Version 58 Local pAUC@0.01 score 0.919106, Leaderboard pAUC@0.01 score 0.9518130 Description: Same blend as sub 44 but increased LGB max_depth from 4 to 5. Worse leaderboard, confirms depth 4 is the sweet spot for LGB.

Submission 49 — random forest 3 test, Version 59 Local pAUC@0.01 score 0.920906, Leaderboard pAUC@0.01 score 0.9547496 Description: Blend weight search with original LGB config (lr=0.05, depth=4, leaves=15). Best local blend was RF=0.65 ET=0.30 LGB=0.05. Much more ET weight than before. New best leaderboard (0.9547). Solo scores: LGB=0.864, RF=0.917, ET=0.926.

